

SAYAN MANDAL

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EDUCATION

Indian Institute of Technology, Guwahati

Bachelor's of Technology in Computer Science and Engineering

2023 – 2027

8/10 CGPA

WORK EXPERIENCE

•Zscaler

May 2026 – Present

Software Development Internship

Hyderabad, India

- Developing ML-driven access recommendation systems for Zscaler Private Access (**ZPA**) to identify and reduce over-permissioned application access using user behavior, transaction logs, and **SCIM group** data.
- Applying clustering techniques such as **K-Means** and **Gaussian Mixture Models (GMM)** on large-scale access and transaction log data to generate access recommendations and improve permission governance.

PROJECTS

•CodeSphere – Online Coding Platform

May 2025 – June 2025

CSEA, IITG

GitHub | Live Site

- Built using **Django**, **Docker**, **AWS EC2**, **AWS ECR**, **Tailwind CSS**, to develop a full-stack **Online Judge** supporting real-time code execution in **C**, **C++**, **Java**, **Python**, and **JavaScript**.
- Integrated **Monaco Editor**, sandboxed code execution with **Docker**, and deployed using **Nginx** and **Gunicorn**.
- Independently developed and maintained the entire platform from backend logic to frontend UI and deployment.
- Enabled hands-on DSA/CP practice for students via a online judge with real-time multi-language execution.

•Internet-Scale BGP Security Analysis using InternetSim

Feb 2026 – Apr 2026

Course Project, IIT Guwahati

GitHub

- Extended an Internet-scale **BGP simulator** (76K+ ASes) and built an interactive **CLI** for real-time simulation.
- Engineered dynamic policy injection to analyze routing behavior under controlled experiments at Internet scale.
- Simulated **BGP prefix hijacking** attacks and evaluated mitigation effectiveness using **Route Origin Validation**.
- Demonstrated that **<1% targeted deployment** secures **~90% network**, reducing hijack impact by over **99%**.

•PyAgent – Autonomous AI Coding Agent

May 2026

Personal Project

GitHub

- Built a CLI autonomous AI coding agent for Python repositories using **Gemini APIs** and **LLM function calling**.
- Designed iterative debugging workflows enabling autonomous repository exploration, bug reproduction, code modification, Python execution, and execution-based fix validation.

•Verizo – Cross-Platform News Application

Jan 2026

Personal Project

GitHub

- Built a production-ready **React Native (Expo)** news app using GNews API with category feeds and topic search.
- Implemented **Firestore Authentication** and **Firestore** to manage user profiles and persist bookmarked articles.
- Used **AsyncStorage** to maintain login sessions and theme preferences (dark/light mode).
- Generated and tested a deployable **Android APK** on real devices; designed for future iOS expansion.

TECHNICAL SKILLS

Web Technologies: HTML, CSS, Tailwind CSS*, Bootstrap, React Native*, Django, REST APIs

Programming: C, C++, Python, JavaScript*, Java*, x86 Assembly (NASM)*

Database Management: MySQL, MongoDB*

Miscellaneous: Docker, Git, GitHub, AWS EC2

ACHIEVEMENTS

- **Codeforces:** Achieved a peak rating of **1423**, earning the title of Specialist; ID: **sayan_mandal** 2025
- **IICPC Codefest 2026 – Prelims:** Ranked **1004/ 9000 participants** (Top 12%) in a national-level CP contest 2026
- **Joint Entrance Examination (Advanced):** Secured **All India Rank 6196** among **0.18 million+ candidates** 2023
- **Joint Entrance Examination (Main):** Secured **All India Rank 4324** among **1.1 million+ candidates** 2023
- **West Bengal Joint Entrance Examination:** Secured **All India Rank 396** among **0.12 million+ candidates** 2023

KEY COURSES TAKEN

- **Computer Science:** Data Structures and Algorithms, Computer Architecture and Organization, Database Management Systems, Design and Analysis of Algorithms, Theory of Computation, Operating Systems, Computer Networks, Randomized Algorithms, Machine Learning, Software Engineering, Cloud Computing, Generative AI for Computer Vision*, Virtual and Augmented Reality*, Software Analysis*.

- **Mathematics:** Linear Algebra, Probability and Random Processes, Number Theory and Abstract Algebra, Optimization, Game Theory

* On-going/ Upcoming